CASEFLOW

Product Datasheet

Version 2.5 | Intelligent Legal Practice Management Platform

August 2026 | Confidential

The Operating System for Indian Legal Practice

AI-Powered | Offline-First | Production-Grade

github.com/AMJATH604/Caseflow_Final

01. Executive Overview

CaseFlow is an AI-powered, offline-first legal practice management platform built specifically for Indian Advocates, Law Firms, and Bar Associations. It serves as the Operating System for Indian Legal Practice - unifying case management, courtroom scheduling, financial ledgers, evidence documentation, AI-assisted legal research, and bar community governance into a single production-grade mobile experience.

Metric	Value
Platform	Android (iOS roadmap Q1 2027)
Stage	Beta - All features free during beta phase
Codebase	80,000+ lines | 36 screens | 40+ components
Architecture	Offline-first, distributed client-server with edge computing
AI Engine	Google Gemini 2.0 Flash (server-side secure proxy)
Target Market	1.7M+ registered advocates in India

02. The Problem

India's 1.7 million registered advocates manage litigation using paper dockets, scattered WhatsApp groups, and unreliable government portals. This creates systemic inefficiencies:

Pain Point	Impact
Missed Hearings	No centralized tracking across courts leading to default orders
Network Dead Zones	Courtrooms have zero connectivity - cloud apps fail completely
Fragmented Tools	Case data scattered across 5-6 separate apps and paper systems
Evidence Integrity	No digital chain-of-custody - evidence easily challenged in court
Research Bottleneck	Manual legal research takes 2-4 hours per case

> *Market Opportunity: India's legal tech market is projected to reach $2.2B by 2028 (NASSCOM). CaseFlow targets the largest underserved segment - individual practitioners and small firms (85% of all advocates).*

03. Core Product Modules

Case & Dossier Management

- CNR-linked case creation with full lifecycle tracking (Filing > Evidence > Arguments > Judgment)
- Case collaboration across firm members with role-based access control
- Sub-modules: Bare Acts, Interim Applications, Interim Orders with PDF preview
- Complete audit trails with soft-delete mechanisms

Court Hearing Calendar & Daily Board

- Date-aware calendar grouped by judicial forum (High Court, District Court, Tribunals)
- 1-tap adjournment workflow with reason tracking and next-date logging
- Real-time Daily Board widget with hearing status flags (active/passed-over/adjourned)

- Automated FCM push reminders via serverless cron job

Financial Ledger & ePay

- Itemized fee agreements, installments, and expense tracking per case
- Auto-generated branded PDF receipts with QR verification
- Court fee ePay integration with smart payment capture

Evidence Lab & Commissioner Module

- GPS geotagged + timestamped evidence photos with SHA-256 cryptographic hashing
- Commissioner toolkit: field visit logs, interactive canvas sketch, witness audio recording
- Full report generator with chain-of-custody verification

AI Legal Assistant (Gemini 2.0 Flash)

- Natural language Q&A across BNS, BNSS, BSA, CPC, Constitution of India
- Zero client-side API key exposure - all calls via secure Supabase Edge Functions
- Context-aware case research and draft generation

Legal Library & Calculators

- Full searchable text of 9 Indian statutes (BNS, BNSS, BSA, CPC, Constitution, etc.)
- 8 interactive calculators: Court Fee, Stamp Duty, Limitation, MACT, Maintenance, and more

Communication Suite

- Chambers Chat: Real-time firm-wide messaging via WebSockets
- Direct Messages: Advocate-to-advocate chat with file attachments
- Broadcast System: Push notices to all firm members instantly

Bar Association Platform (BarDesk)

- Digital membership cards with QR verification, election voting, public notices

Citizen Portal

- Case status tracking for litigants, Know Your Rights guide, legal tools access

04. Technology Stack

Layer	Technology	Rationale
Mobile	Flutter 3.29 / Dart 3.7	Single codebase, 60 FPS native
State Mgmt	Riverpod	Type-safe, compile-time checked
Routing	GoRouter 14.x	Declarative nav with auth guards
Offline DB	SQLite (sqflite)	Full offline data access
Backend	Supabase (PostgreSQL 15)	Managed infra, RLS, auth, realtime
Auth	PKCE + JWT + OAuth 2.0	Industry-standard, token rotation
AI Engine	Google Gemini 2.0 Flash	Low-latency legal NLP, server proxy
PDF Engine	iLovePDF API (proxied)	Compress, merge, convert, OCR
Storage	Cloudflare R2	S3-compatible, zero egress cost
Push	Firebase Cloud Messaging	V1 HTTP API, topic-based routing

CI/CD

GitHub Actions

Auto-build signed APK on push

Responsive

flutter_screenutil

390x844 baseline, 320-430dp+

05. System Architecture

CaseFlow employs a Distributed Client-Server & Edge Architecture. The client is a reactive Flutter application with Riverpod state management and SQLite offline storage. The backend leverages Supabase (PostgreSQL, Realtime, Auth) with Deno Edge Functions and Cloudflare R2.

Layer	Components	Responsibilities
Client (Mobile)	Flutter 3.29, Riverpod, SQLite	UI, offline caching, state mgmt
Client (Web)	React 18, Vite	Admin dashboard, analytics
API Gateway	Supabase PostgREST, Realtime	REST APIs, WebSocket streams
Database	PostgreSQL 15, RLS	Persistence, row-level security
Edge Functions	Deno Serverless (3 functions)	Gemini proxy, PDF proxy, FCM cron
Storage	Cloudflare R2 + Supabase	Documents, evidence, attachments
Notifications	Firebase Cloud Messaging	Push reminders, hearing alerts

06. Security & Compliance

Security Feature	Implementation
Row Level Security	Enforced on 100% of database tables
API Key Isolation	Gemini & PDF keys in Edge Function secrets - never in client
Evidence Integrity	SHA-256 cryptographic hashing on all evidence photos
Authentication	PKCE flow with refresh token rotation
Data Isolation	Firm-visibility OR-filter - own + firm data only
Encrypted Storage	At-rest encryption on all backend storage
Audit Logging	Full operation audit trail with user attribution
Architecture	ISO/IEC 27001 compliant design patterns

07. Performance Benchmarks

Metric	Target	Implementation
Cold Start	< 2.0 seconds	Riverpod lazy initialization
Screen Navigation	< 100ms	State caching + pre-built routes
Scroll Performance	60 FPS	Paginated lists, cached images
Dashboard Load	~3x faster	Future.wait() parallel queries
Offline Access	100% reads	SQLite-backed offline mode
Push Delivery	< 5 seconds	FCM V1 HTTP API + topic routing
Uptime	99.9%	Supabase managed + Cloudflare

08. User Personas & Market Fit

Persona	Pain Points Solved	Success Metrics
Senior Advocate	Firm oversight, revenue tracking	< 2s daily oversight per case
Junior Advocate	Research bottleneck, diary chaos	50% faster research & drafting
Court Clerk	Filing deadlines, doc management	0 missed filing deadlines
Citizen Litigant	Case opacity, legal awareness	Transparent real-time updates

09. Competitive Differentiation

Feature	CaseFlow	Traditional Apps
Offline-First	Full SQLite cache - works in court	Requires internet
AI Legal Research	Gemini 2.0 Flash + secure proxy	No AI integration
Evidence Hashing	SHA-256 chain-of-custody	No integrity verification
Real-time Chat	WebSocket chambers + DMs	Relies on WhatsApp
Bar Association	Full governance platform	Not covered
Citizen Portal	Case tracking + Know Your Rights	Advocate-only
Indian Legal Focus	BNS/BNSS/BSA/CPC native	Generic or Western-focused

10. Product Scale

Dimension	Current State
Codebase	80,000+ lines of production Dart/Flutter code
Screens	36 fully functional screens
Components	40+ reusable widget library
Database Tables	15+ with full RLS policies
Edge Functions	3 serverless functions (AI, PDF, Push)
Legal Content	9 Indian statutes fully digitized
Calculators	8 interactive legal calculators
CI/CD	Fully automated push-to-APK pipeline

11. Roadmap

Timeline	Milestone
Q3 2026	Public Beta launch on Google Play Store
Q4 2026	Bar Association pilot programs (3-5 associations)

Q1 2027

iOS release, advanced analytics dashboard

Q2 2027

eCourts API deep integration, multi-language support

Q3 2027

SaaS monetization - Professional & Enterprise tiers

12. Team & Contact

Role	Name
Founder & Lead Developer	Amjath
Mentored By	Ananthan Ayyasamy

Repository: github.com/AMJATH604/Caseflow_Final

A. Appendix - Documentation Suite

For deep-dive technical details, the following documents are available on request:

#	Document	Contents
1	Product Requirement Document	User personas, feature specs, RICE priority
2	Technical Requirement Document	System architecture, data flow, API specs
3	App Flow & UI/UX Design Brief	Screen navigation, interaction patterns
4	Backend Schema Specification	PostgreSQL tables, RLS policies
5	Implementation Chronology	Sprint-by-sprint build timeline
6	README	Quick start, tech stack, CI/CD pipeline

CaseFlow - The Operating System for Indian Legal Practice

Built with love for the Indian legal community